

Predictive Modeling of Residential Property Prices in Pierce County

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Preface

This documentation presents the project proposal and development progress for the Predictive Modeling of Residential Property Prices in Pierce County, developed as part of the Solution Engineering course at FHTW.

The project aims to build a robust machine learning pipeline to estimate housing prices by integrating diverse data sources from the Pierce County Assessor-Treasurer. This book serves as the central repository for project objectives, data descriptions, organizational structure, and development plans.

Project Team

- **Mira Radakovic** (Project Leader)
- **Edberto Moura Lima** (Data Analyst)
- **Gerta Hajrullahi** (Data Analyst)
- **Helga Deutschmann** (Data Analyst)

Document Status

This version corresponds to the **Pr1oject Proposal (Report 1)**.

- ☒ Project Objectives (Qualitative & Quantitative)
- ☒ Data Source Description & Dictionary
- ☒ Project Organization & Task Allocation
- ☒ Development Plan & Workflow
- ☒ Gantt Schedule & Milestones

1 Project Objective

1.1 Background & Motivation

Estimating house prices is a classic regression problem with clear real-world importance. Housing markets are influenced by many interconnected factors, including property characteristics, location, neighborhood effects, and market trends over time. Because nearby properties often share similar values and market conditions change continuously, house-price prediction provides an excellent example for applying the complete data-science workflow covered in this course.

For this project, the publicly available Assessor–Treasurer data from Pierce County is used as the main data source. The dataset contains detailed and realistic information about properties, transactions, and assessments across multiple linked tables, offering a large-scale and practical environment for data exploration, pre-processing, feature engineering, and predictive modeling.

1.2 Project Goal

Develop a machine learning system capable of predicting residential property prices using structural, spatial, economic, and neighborhood characteristics.

1.2.1 Qualitative Objectives

1. Build a clean housing dataset for Pierce County.
2. Explore which factors influence house prices.
3. Engineer predictive variables from property and spatial data.
4. Compare traditional statistical and machine learning approaches.
5. Deploy a simple Shiny app for interactive price estimation.
6. Allow non-technical users to test predictions using custom property inputs.
7. Create visual outputs that support decision-making.

1.3 Quantitative Objectives

Objective	Metric	Target
Model Accuracy	Root Mean Square Error (RMSE)	< USD 75,000
Explanatory Power	Coefficient of Determination (R^2)	> 0.85
Model Diversity	Algorithm Benchmarking	4 distinct algorithms
Feature Richness	Engineered Predictors	20 model-ready features
App Performance	Real-time Inference	Response time < 3 seconds
Data Integrity	Table Integration	100% of selected source tables joined

1.4 Deliverables & Timeline

- **21.05.2026:** Project Proposal Presentation
- **06.06.2026:** Interim Results Presentation
- **24.06.2026:** Final Results Presentation & Project Report Submission
- **24.06.2026:** Live demo of deployed model App

2 Data Source Description & Data Dictionary

2.1 Overview

The Pierce County Assessor-Treasurer Data Mart contains comprehensive property assessment and appraisal data maintained by Pierce County, Washington. The data is organized into multiple related tables that track property characteristics, improvements, valuations, and sales transactions.

This data dictionary is a prerequisite for data preparation and serves as the reference for understanding the relational structure and field definitions necessary for proper data integration and feature engineering.

2.2 Data Sources

- **Tax and Assessment System:** Maintains current tax account information
- **Appraisal System:** Maintains detailed property appraisals and improvements
- **Public Records:** Available through Pierce County Public Records Request Center

Note

RCW 42.56.070(9) may restrict the release of taxpayer names and mailing addresses in certain circumstances. Contact the Pierce County Public Records Request Center for additional information.

2.3 Data Mart Architecture

2.3.1 Table Relationships

The data mart uses a relational schema with the following structure:

Tax Account (Parcel Number)

↓

Appraisal Account (Parcel Number)

Tax Description (Parcel Number, Line Number)

Land Attribute (Parcel Number, Attribute)

Improvement (Parcel Number, Building ID)

Improvement Built-As (Parcel Number, Building ID, Built-As Number)

Improvement Detail (Parcel Number, Building ID, Detail Type)

Seg/Merge (Seg/Merge Number, Parcel Number)

Sale (ETN, Parcel Number)

Taxpayer (Parcel Number)

💡 Key

Field names in **bold** represent primary key(s).

2.4 Table Definitions

2.4.1 Tax Account Table

Tracks current tax and assessment account information for each property.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Account Type	Text	5	REAL - Real Property; STRUC - Structures; PERS - Personal Property; MOBIL - Mobile Home

Field Name	Type	Size	Description
Property Type	Text	5	ASIMP - Administrative Seg Improvement; LNDIM - Land and Improvements; MBLHM - Mobile Home; RRLEA - Railroad Leases; SAPP - State Assessed Personal Property; SARP - State Assessed Real Property; STRUC - Leased Land and/or Structure
Site Address	Text	50	Physical location address of the property. May be different than mailing address. "XXX" indicates estimated address using GIS
Use Code	Text	5	4-digit numeric code for present highest and best use of the property
Use Description	Text	50	Text description of the numeric use code
Tax Year - Prior	Integer		Tax year for which all prior year values apply
Tax Code Area - Prior Year	Text	10	Code describing geographical area and specific combination of taxing districts. Subject to annual change
Exemption Type - Prior Year	Text	40	Type of exemption, if any, applied to the property

Field Name	Type	Size	Description
Current Use Code - Prior Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Prior Year	Integer		Total fair market value of land, as determined by appraisal process
Improvement Value - Prior Year	Integer		Total fair market value of improvements (buildings, structures), as determined by appraisal process
Total Market Value - Prior Year	Integer		Total fair market value of land and improvements, as determined by appraisal process
Taxable Value - Prior Year	Integer		Market value of the property, minus any exemptions granted
Tax Year - Current	Integer		Tax year for which all current year values apply
Tax Code Area - Current Year	Text	10	Code describing geographical area and specific combination of taxing districts
Exemption Type - Current Year	Text	40	Type of exemption, if any, applied to the property

Field Name	Type	Size	Description
Current Use Code - Current Year	Text	5	AGRI - Agricultural; CT - Current Use Timber; FORDG - Forest Designated; OPBRS - Open Space PBRS; OPEN - Open Space
Land Value - Current Year	Integer		Total fair market value of land for the parcel
Improvement Value - Current Year	Integer		Total fair market value of improvements (buildings, structures) for the parcel
Total Market Value - Current Year	Integer		Total fair market value of land and improvements for the parcel
Taxable Value - Current Year	Integer		Market value of the property, minus any exemptions granted
Range	Text	10	Range
Township	Text	10	Township
Section	Text	20	Section
Quarter Section	Text	20	Quarter
Subdivision Name	Text	50	Subdivision name
Located On Parcel	Text	10	Real property parcel number on which this improvement-only parcel is located (Personal Property, Mobile Homes, Leasehold Structures)

2.4.2 Appraisal Account Table

Detailed property appraisal data with characteristics and valuations.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Appraisal Account Type	Text	15	Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest
Business Name	Text	50	Main business name on the parcel, or space number for mobile/manufactured home in a park
Value Area ID	Text	10	Identifies the Physical Inspection cycle for each parcel (PI1 to PI6)
Land Economic Area	Text	30	Also known as LEA, ties parcels to land valuation models. For Commercial: identifies land use model by PI year. For Residential: location code identifying appraisal area, sector and subsector. For Res Com Condos: relates directly to account land value
Buildings	Integer		Represents the number of improvements (buildings) located on a parcel

Field Name	Type	Size	Description
Group Account Number	Text	30	Identifies and ties all parcels qualifying as one economic unit (contiguous assessment). Also identifies all parcels in a condominium project
Land Gross Acres	Decimal	15.4	Total acres that make up the economic unit which could include multiple parcels
Land Net Acres	Decimal	15.4	Size of the individual parcel in acres
Land Gross Square Feet	Decimal	15.4	Total square feet that make up the economic unit which could include multiple parcels
Land Net Square Feet	Decimal	15.4	Size of the individual parcel in square feet
Land Gross Front Feet	Decimal	15.4	For commercial parcels: front feet of the site facing the arterial. For residential: front feet of the lot facing the address side. For waterfront: effective waterfront length of individual lot or combination of lots under single ownership

Field Name	Type	Size	Description
Land Width	Integer		Residential Use Only. Equal to the Net Effective Waterfront Front Feet of the individual parcel, or using Group Account allocation method, then the sum of Net Effective Waterfront Front Feet for the group
Land Depth	Integer		Residential Use Only. Equal to the Net Effective Depth of the individual waterfront parcel, or using Group Account allocation method, then the sum of Net Effective Depths for the waterfront group. Measured from the top of the bank (where bulkhead begins) to the inland property line
Submerged Area Square Feet	Integer		Area of a parcel that is submerged. Not included in the Gross or Net Square Feet
Appraisal Date	Date		Indicates the most recent date an appraiser physically observed the property and/or made changes to the record
Waterfront Type	Text	30	Describes the type of waterfront the property adjoins or has legal access to

Field Name	Type	Size	Description
View Quality	Text	30	Assigned to reflect the market appeal of the overall view available from the dwelling or property
Utility Electric	Text	30	Identifies if power is installed, available or is not available on the property
Utility Sewer	Text	30	Identifies if sewer/septic is installed, available or not available if the property does not support an on site sewage disposal system (no perc)
Utility Water	Text	30	Identifies if water is installed, available or is not available
Street Type	Text	30	Identifies if the access street is paved or unpaved. If blank, the parcel could be a reference parcel or a mobile home assessed as personal property
Latitude	Decimal	10.5	The latitude of the parcel centroid. Some parcels may not have latitude/longitude coordinates (building-only, mineral rights and reference parcels, as well as others not represented in the GIS system)
Longitude	Decimal	10.5	The longitude of the parcel centroid

2.4.3 Improvement Table

Detailed building improvement data linked to appraisal accounts.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel. Not all improvements are on a separate record and may not always be in sequence
Property Type	Text	15	Links buildings to proper cost and depreciation tables. Options include: Commercial, Duplex, Industrial, Mobile Home, Multiple Unit, Out Building, Residential, Townhouse, Triplex

Field Name	Type	Size	Description
Neighborhood	Text	10	Represents boundaries for land subject to similar social, environmental, economic and governmental forces. Boundaries are typically located within a geographic area defined by natural, man-made, or political boundaries. Residential Neighborhoods and LEA's are the same. Commercial Neighborhoods define the location of the property while Commercial LEA's define the use of the property

Field Name	Type	Size	Description
Neighborhood Extension	Text	10	Based on property type: Commercial - represents the primary occupancy or use of the property. Residential - not used, therefore they are set to zero. Mobile/Manufactured Homes - based upon the underlying land ownership. All Mobile/Manufactured homes located on land owned by the Mobile/Manufactured home owner are coded with the Neighborhood Extension 0. Exception - Mobile/Manufactured homes located within a park and owned by the park owner, will carry the same Neighborhood Extension as the park. Mobile Homes in a park have a neighborhood extension of: PL 1 (P = Park, L = Leased); P2, P3, P4, etc

Field Name	Type	Size	Description
Square Feet	Integer		Sum of the square feet for the 'built as' types identified for the building. Each building may have one or more 'built as' type (see IMPROVEMENT BUILTAS table), though usually only one. The exceptions are 'built as' # 124 and 126 (Add on only Res & Com) or any type of Storage Tank, where it represents the count of those 'built as' as opposed to the sum of their square feet
Net Square Feet	Integer		Used primarily to calculate income on commercial properties, where it typically represents the sum of the net rentable area(s). This number may be representative of one or more buildings. If part of a Group Account, these buildings may be located on more than one parcel. For Residential it is equal to the built as square feet or zero (0)

2.4.4 Improvement Built-As Table

Detailed specifications for building construction styles and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Built-As Number	Integer		Unique number identifying each Built-As on an improvement. Sequential number assigned by the system starting with the largest Built-As being number 1
Built-As ID	Integer		Number associated with the description of the purpose or style of construction of the building
Built-As Description	Text	255	Original purpose for and/or style of construction of the building. The Built As is associated with the Marshall and Swift cost and depreciation tables

Field Name	Type	Size	Description
Built-As Square Feet	Integer		Sum of the square feet for the 'built as' types identified for the building. Each building may have one or more 'built as' type, though usually only one. The exceptions are 'built as' # 124 and 126 (Add on only Res & Com) or any type of Storage Tank
HVAC	Integer		Code associated with the predominant heating source for the built-as structure
HVAC Description	Text	30	Text description associated with the predominant heating source for the built-as structure (Forced Air, Electric Baseboard, Steam, etc)
Exterior	Text	25	Predominant type of construction materials used for the exterior siding on Residential Buildings
Interior	Text	15	Predominant type of materials used on the interior walls (Sheetrock or Paneling). Collected for Residential and Mobile Homes only
Stories	Decimal	13.2	Number of floors/building levels above grade. Stories do not include attic or basement areas

Field Name	Type	Size	Description
Story Height	Integer		Based on property type: Commercial - determined by taking an average of the overall ceiling heights of all floors for the Blt As. Residential - determined by the majority of the story height for the first floor (main). Story height is automatically rounded by the system (The system rounds up for .5 and higher and rounds down for anything below .5)
Sprinkler Square Feet	Integer		Total square footage covered by a sprinkler system. Sprinkler SF does not always equal the built as square footage
Roof Cover	Text	20	Material used for the roof (Composition Shingles, Wood Shake, Concrete Tile, etc)
Bedrooms	Integer		Number of bedrooms listed for a residential property (Collected for informational purposes only)

Field Name	Type	Size	Description
Bathrooms	Decimal	7.2	Number of baths listed for a residential property. The number is listed as a decimal, i.e. 2.75 = two full and one three-quarter baths. A tub/sink/toilet combination (plus any additional fixtures) is considered 1.0 bath. A shower/sink/toilet combination (plus any additional fixtures) is 0.75 bath. A sink/toilet combination is .5 bath
Units	Integer		Number of separate units within the building. For Commercial properties, Units may be combined to generate an income approach for the economic unit on building one
Class Code	Text	20	Code for one of five basic cost groups by type of framing (supporting beams and columns), walls, floors and roof structures, and fireproofing. Typically Commercial Buildings

Field Name	Type	Size	Description
Class Description	Text	50	Text description for one of five basic cost groups by type of framing (supporting beams and columns), walls, floors and roof structures, and fireproofing. Typically Commercial Buildings
Year Built	Integer		Year the building was built, as stated by the building permit or a historical record
Year Remodeled	Integer		Year in which improvements are made to the existing structure that extend the life of the structure
Adjusted Year Built	Integer		If greater than the Year Built, it indicates improvements have been made to the original property over and above normal maintenance which would effectively reduce the age of the building
Physical Age	Integer		Tax year less the adjusted year built equals the physical age of the built as
Built-As Length	Integer		Length of the mobile/manufactured home

Field Name	Type	Size	Description
Built-As Width	Integer		Width of the mobile/manufactured home
Mobile Home Model	Text	50	Model of the mobile/manufactured home

2.4.5 Improvement Detail Table

Additional specific improvement characteristics and features.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Building ID	Text	5	Unique identifier for each improvement record on a parcel
Detail Type	Text	10	Category that specific improvement characteristics are coded under. Options include: Add On, Appliance, Balcony, Basement, Carport, Elevator, Fixture, Garage, Mezzanine, Porch, Rough In, Storage. The Add On type allows the appraiser to assign a quality different from the main structure for the characteristics within this field

Field Name	Type	Size	Description
Detail Description	Text	50	Brief explanation of the improvement characteristic (Fireplace, PreFab/Stoves, Laundry Facility, Type of Porch (Cvrd Wd Deck), Type of Bath (Bath 2 Fixture))
Units	Decimal	15.4	Component parts of the building characteristics. The units indicate either the square footage of the structure or a count of each component. Examples: Appliance, Allowance, 1 (count) or Porch, Open Slab, 337 (sf), or Fixture – Bath 3 Fixtures – 2 (count)

2.4.6 Land Attribute Table

Land-specific attributes and characteristics affecting valuation.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property

Field Name	Type	Size	Description
Attribute	Text	30	Describes the category that a land attribute is assigned to: Residential (R), Commercial (C) or Res Com Condo (X) and the subcategory, such as Amenities, Economic, Functional, Neighborhood, Utilities, etc
Attribute Description	Text	30	Further defines the land attribute. Land attributes may or may not contribute to value (Examples of attributes: View Avg, NBHD Gated, WF Bank Low, water available etc)

2.4.7 Sale Table

Historical property sales transaction data.

Field Name	Type	Size	Description
ETN	Text	11	Recording number issued by the Auditor's Office when a property transaction is recorded. Excise tax must be paid before the document of conveyance can be recorded or the mobile home title transferred
Parcel Count	Integer		Number of parcels associated with a sale
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Sale Date	Date		Date the legal document (deed) was executed
Sale Price	Decimal	15.2	Dollar amount recorded on the ETN
Deed Type	Text	40	Type of document which conveyed an interest in or legal title to the property
Grantor	Text	80	Individual(s) or company conveying ownership or interest in the property described on the deed
Grantee	Text	80	Individual(s) or company purchasing ownership or interest in the property described on the deed

Field Name	Type	Size	Description
Valid/Invalid	Text	7	Used to code the validity of a sale for appraisal purposes. A sale in the open market between two unrelated parties, each of whom is reasonably knowledgeable of market conditions and under no undue pressure to buy or sell would be considered a valid transaction. However, a sale can be a valid transaction but if the conditions of the sale fit one of the 27 deletion categories identified in the State Ratio RCW, the sale would be coded invalid for Assessor purposes (sale between relatives, estate sale, percent interest, etc)
Confirmed/Unconfirmed	Text	11	Identifies if the property characteristics at the time of the sale and/or circumstances of the sale were verified by Assessor-Treasurer staff

Field Name	Type	Size	Description
Exclude Reason	Text	30	State required description of why a sales transaction must be excluded from sales studies. Examples include non-arms length sales transactions such as “Family different last names”, sale transactions with undue pressure such as “Estate sale” and “Forced Sale Trans in Lieu Fcl”, and sales where the property characteristics have changed since the sale, “Improved after sale”
Improved/Vacant	Text	8	If any parcel in the sale has any building value it is coded as improved. Vacant is vacant land with no building value
Appraisal Account Type	Text	15	How a parcel is classified (Com Condo, Com Leasehold, Com Multi Unit, Commercial, Industrial, Mobile Home, Reference, Res Com Condo, Res Leasehold, Residential, Trended Invest)

2.4.8 Seg/Merge Table

Property segregation and merger transaction history.

Field Name	Type	Size	Description
Seg/Merge Number	Text	12	Unique number assigned to each segregation
Parent/Child Indicator	Text	1	P - Parent Parcel - inactivated parcel being replaced by the new child parcel. There may be zero, one or more parent parcels involved with each segregation. In some cases, a parent parcel may remain active after the segregation. C - Child Parcel - a new parcel created as a result of the seg. There are one or more child(ren) parcels involved with each segregation
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Continued Indicator	Text	1	Y - the parcel is still active after the segregation; N - the parcel is no longer active after the segregation
Completed Date	Date		Date that the segregation was completed
Tax Year	Text	4	Tax year of the segregation

2.4.9 Taxpayer Table

Property owner information (subject to privacy restrictions).

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property

Warning

RCW 42.56.070(9) may restrict the release of taxpayer names and mailing addresses in certain circumstances. Please contact the Pierce County Public Records Request Center to obtain additional information and/or initiate a request for the taxpayer table.

2.4.10 Tax Description Table

Detailed tax coding and descriptions for parcels.

Field Name	Type	Size	Description
Parcel Number	Text	10	Unique 10 digit number assigned to each property
Line Number	Integer		Sequential number identifying each line of a tax description
Tax Description Line	Text	200	Tax description text

2.5 Key Relationships

- **One-to-Many:** Each parcel (Parcel Number) can have multiple improvements (Building IDs)
 - **One-to-Many:** Each improvement can have multiple built-as specifications and detail records
 - **One-to-One:** Each parcel has one appraisal account and one tax account
 - **One-to-Many:** Each parcel can have multiple sales transactions (ETN)
 - **One-to-Many:** Each parcel can have multiple land attributes
 - **One-to-Many:** Each parcel can be involved in multiple seg/merge transactions
-

2.6 Data Quality Notes

- **Historical Data:** Seg/Merge and Sale tables contain historical data. All other tables contain data for active parcels only
 - **Missing Coordinates:** Some parcels (building-only, mineral rights, reference parcels) may not have latitude/longitude coordinates
 - **Address Estimation:** If address contains “XXX”, it was estimated using GIS
 - **Appraisal Dates:** Fields like Appraisal Date indicate when the property was last physically inspected
 - **Value Dates:** Land, Improvement, and Total Market Values are tracked for both Prior Year and Current Year
-

2.7 Data Dictionary Notes

- **Field names in bold** represent primary key(s) in their respective tables
 - **Size column** represents maximum field length for Text and Decimal fields
 - Fields without size notation are fixed or variable based on actual values
 - Most text fields are case-sensitive as recorded in the assessor system
-

2.8 Access & Contact

For additional information, data downloads, or to request specific datasets:

- **Source:** Pierce County Assessor-Treasurer Office
- **Website:** <https://www.piercecountywa.gov/>
- **Records Request:** Pierce County Public Records Request Center

3 Project Organization & Task Allocation

3.1 Group Structure

Name	Role
Mira Radakovic (PL)	Project Leader
Edberto Moura Lima	Data Analyst
Gerta Hajrullahi	Data Analyst
Helga Deutschmann	Data Analyst

3.2 Tools & Infrastructure

Component	Detail
Language	R (primary)
Database	Supabase (PostgreSQL)
Version Control	Git + GitHub
Project Management	GitHub Projects (Kanban Board)
Communication	Group messaging + sync meetings
Documentation	Quarto
Data Wrangling	dplyr, tidyr, readr
EDA & Viz	ggplot2, corrplot, leaflet, plotly
Modelling	caret / tidymodels, ranger, xgboost, glmnet
Spatial Analysis	sf, geosphere
Deployment	Shiny, shinydashboard

3.3 Project Kanban Board

The following board tracks the real-time status of all project tasks.



3.4 Risk Management

Risk	Mitigation
Large number of tables / complex joins	Build join pipeline incrementally; validate row counts at each step
Many missing values in building details	Imputation strategies (median/mode); sensitivity analysis on imputed variables
Data covers multiple property types (commercial, mobile homes)	Filter to Residential Improved sales only at ingestion
Temporal drift in prices (market cycles)	Include sale year/month as features; optionally restrict to recent years
Skewed Sale Price distribution	Log-transform target; evaluate models on both scales
Shiny App complexity	Start with minimal viable app; add features iteratively

4 Development Plan Overview

The project is splinted into work packages (WP0–WP7) over 6 weeks.

Detailed work package specifications (tasks, deliverables, quality criteria, and dependencies) are provided in [Chapter 5: Project Schedule & Work Packages](#).

4.1 Workflow Diagram

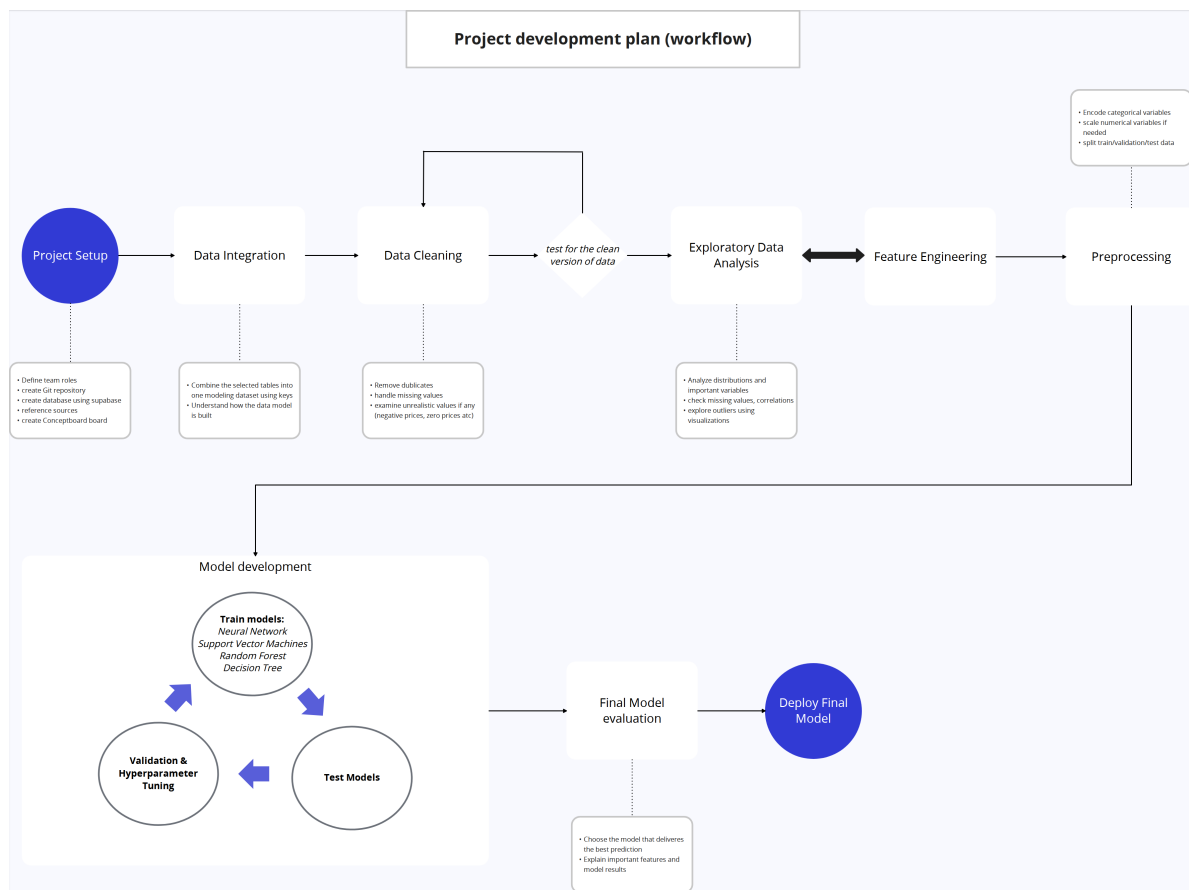


Figure 4.1: Project Development Workflow

4.2 Work Package Summary

WP	Phase	Duration
WP0	Project Setup	13–19 May
WP1	Data Integration	20–26 May
WP2	Data Cleaning	27 May – 2 June
WP3	EDA & Features	30 May – 9 June
WP4	Preprocessing	7–9 June
WP5	Model Development	9–15 June
WP6	Model Evaluation	16–20 June
WP7	Deployment	20–24 June

4.3 Communication & Oversight

Weekly Sync: - Review completed tasks - Discuss blockers & dependencies - Update Kanban board - Plan next week

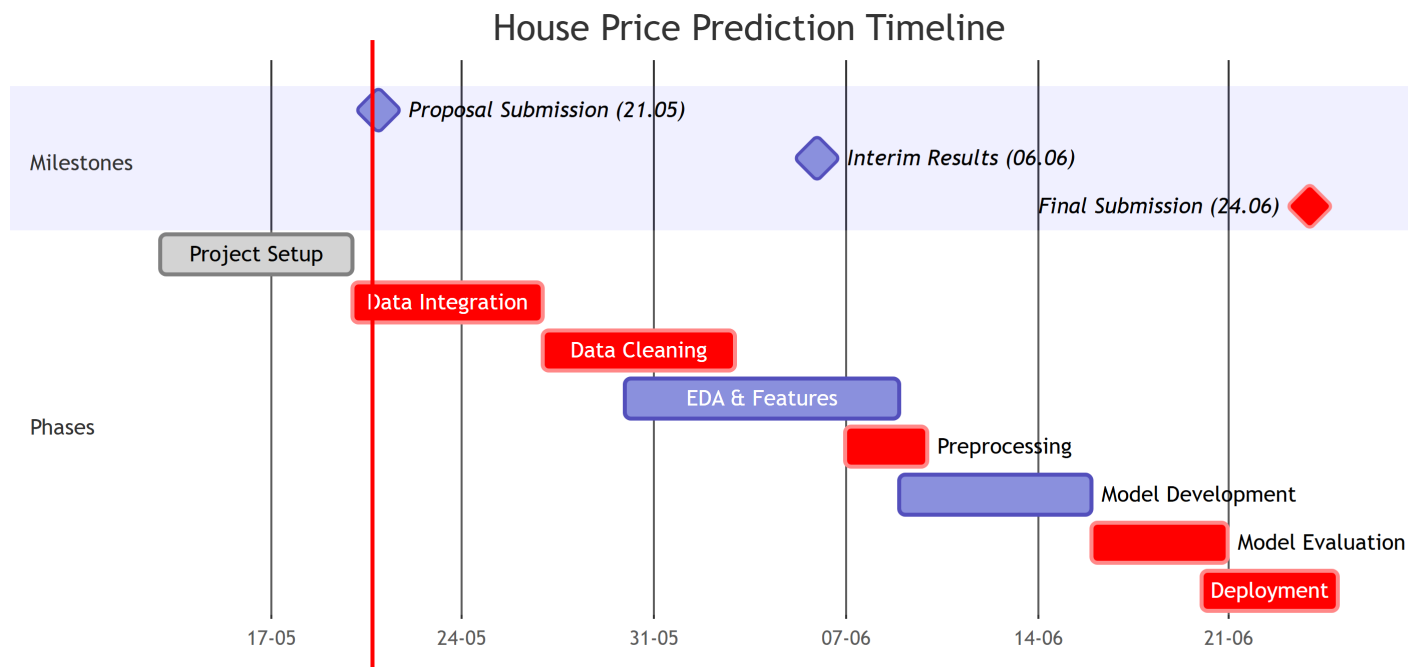
Status Reporting: Mira provides weekly status to lecturer

5 Project Schedule & Work Packages

Project Duration: 13 May – 24 June 2026 (6 weeks)

5.1 Interactive Gantt Chart

The following chart outlines the project timeline, milestones, and critical path.



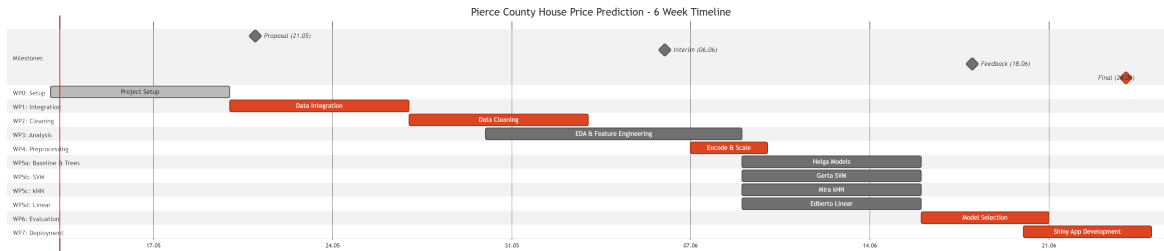


Figure 5.1: Project Gantt Chart (Static Version)

5.2 Detailed Work Package Definitions

5.2.1 WP0: Project Setup

Duration: 13–19 May | **Status:** Done

5.2.1.1 Tasks

- Define team roles and responsibilities
- Create Git repository (private, shared with lecturer)
- Set up Supabase database connection
- Initialize Kanban board (Backlog/To Do/Doing/Testing/Done)
- Configure R development environment

5.2.1.2 Deliverables

GitHub repository Supabase connection Kanban board Team charter document

5.2.2 WP1: Data Integration

Duration: 20–26 May | **Status:** To Do

5.2.2.1 Tasks

1. Download Pierce County data files
2. Map table relationships and join keys
3. Perform sequential table joins (validate at each step)
4. Validate referential integrity and row counts
5. Create data integration report

5.2.2.2 Deliverables

- Integrated dataset (.RData)
- Table mapping diagram
- Data dictionary (updated)
- Validation report with row counts

5.2.2.3 Quality Criteria

- ☐ All tables joined successfully
 - ☐ Row counts documented at each step
 - ☐ No unexpected NULL values
 - ☐ Referential integrity verified
-

5.2.3 WP2: Data Cleaning & Preparation

Duration: 27 May – 2 June | **Status:** To Do

5.2.3.1 Tasks

1. Filter to Residential Improved properties only
2. Remove duplicate records
3. Handle missing values (document strategy)
4. Examine and document outliers
5. Variable selection — keep only analysis-ready columns
6. Run validation test suite

5.2.3.2 Deliverables

- Cleaned dataset (.RData)
- Data cleaning log
- Missing value analysis
- Quality assurance report

5.2.3.3 Quality Criteria

- ☐ No critical missing values
 - ☐ Data types verified
 - ☐ Outliers documented
 - ☐ Test suite passes
-

5.2.4 WP3: Exploratory Data Analysis & Feature Engineering

Duration: 30 May – 9 June | **Status:** To Do

5.2.4.1 WP3a: EDA Tasks

- Univariate analysis (distributions, summaries)
- Bivariate analysis (price vs. key variables)
- Spatial analysis with leaflet maps
- Correlation matrix and hypothesis testing

5.2.4.2 WP3b: Feature Engineering Tasks

- Create derived features (age, price/sqft, neighborhood effects, etc.)
- Encode categorical variables
- Feature selection — target 15 final features
- Document feature importance baseline

5.2.4.3 Deliverables

- EDA report with visualizations
- Correlation heatmap
- Feature engineering log
- Final feature set documentation (15 features)

5.2.4.4 Quality Criteria

- ☐ All visualizations publication-ready
 - ☐ 15 features engineered and named
 - ☐ No data leakage from target variable
 - ☐ Feature-target correlations documented
-

5.2.5 WP4: Preprocessing

Duration: 7–9 June | **Status:** To Do

5.2.5.1 Tasks

1. Encode categorical variables (finalize encoding scheme)
2. Scale numerical variables (standardize or normalize)
3. Train/validation/test split (70/15/15% stratified)
4. Document random seed for reproducibility

5.2.5.2 Deliverables

- 3 split datasets (train, validation, test)
- Encoding mapping tables
- Scaling parameters (saved as R object)
- Data split documentation

5.2.5.3 Quality Criteria

- ☐ No data leakage (scaling only from training set)
 - ☐ Balanced distribution across splits
 - ☐ Random seed documented
 - ☐ All features properly encoded/scaled
-

5.2.6 WP5: Model Development (Parallel Tracks)

Duration: 9–15 June | **Status:** To Do

5.2.6.1 WP5a: Baseline & Tree Models

- Baseline Linear Regression
- Decision Tree (hyperparameter tuning)
- Random Forest (grid search over parameters)

5.2.6.2 WP5b: Support Vector Machine

- SVM setup (kernel selection)
- Hyperparameter tuning (C, epsilon)
- Cross-validation

5.2.6.3 WP5c: k-Nearest Neighbours

- Setup (distance metric, k selection)
- kNN based on: distance (lon, lat) + square metre
- Hyperparameter optimization

5.2.6.4 WP5d: Linear Regression

- Full feature set training
- Feature importance extraction
- Residual diagnostics

5.2.6.5 Deliverables

- Trained model objects (.RDS)
- Cross-validation results
- Validation & test metrics
- Model configuration documentation

5.2.6.6 Quality Criteria

- ☐ All 5 models trained and saved
- ☐ Cross-validation RMSE < 100k (preliminary)
- ☐ No data leakage to validation/test
- ☐ Test predictions generated for all samples

5.2.7 WP6: Model Evaluation & Selection

Duration: 16–20 June | **Status:** To Do

5.2.7.1 Tasks

1. Cross-validation for all 5 models (k=5 fold)
2. Performance benchmarking (RMSE, MAE, R^2)
3. Create model comparison table and visualizations
4. Select best model based on criteria:
 - $RMSE < USD\ 80,000$
 - $R^2 > 0.80$
5. Extract and visualize feature importance

5.2.7.2 Deliverables

- Model comparison table (CSV)
- Benchmarking plots (PNG/PDF)
- Best model selection report
- Feature importance ranking
- Final model object saved

5.2.7.3 Quality Criteria

- ☐ All models evaluated on same test set
 - ☐ Best model meets $RMSE < 80k$ & $R^2 > 0.80$ targets
 - ☐ Feature importance interpreted
 - ☐ Selection criteria documented
-

5.2.8 WP7: Deployment & Documentation

Duration: 20–24 June | **Status:** To Do

5.2.8.1 Tasks

1. Build Shiny App (UI + server logic)
2. Create prediction endpoint (returns estimate + confidence interval)
3. Add feature importance visualization
4. Final validation testing
5. Prepare presentation slides
6. Assemble final report
7. Deploy to shinyapps.io or local server

5.2.8.2 Deliverables

- Deployed Shiny App
- App source code (app.R)
- Presentation slides (PDF)
- Final report (PDF)
- User guide & model documentation
- All code on GitHub

5.2.8.3 Quality Criteria

- ☐ Shiny App loads without errors
 - ☐ Predictions return in <5 seconds
 - ☐ Confidence intervals displayed
 - ☐ Presentation covers all key findings
 - ☐ Report submitted via Moodle by deadline
-

5.3 Dependencies & Critical Path

WP	Depends On	Notes
WP1	WP0	Requires Git repo & Supabase setup
WP2	WP1	Requires integrated dataset
WP3	WP2	Requires clean dataset
WP4	WP3	Requires feature set
WP5	WP4	Requires preprocessed data
WP6	WP5	Requires all trained models

WP	Depends On	Notes
WP7	WP6	Requires selected best model

Critical Path Tasks: 1. WP1: Data integration (26.05) 2. WP2: Data cleaning (02.06) 3. WP3: Feature engineering (09.06) 4. WP4: Train/test split (09.06) 5. WP5: Model training (15.06) 6. WP6: Model selection (20.06) 7. WP7: Shiny App deployment (23.06)

5.4 Key Milestones

Date	Event	Deliverables
21.05.2026	Proposal Presentation	Objectives, data dictionary, organization, development plan
06.06.2026	Interim Results	Data status, EDA findings, initial model results
18.06.2026	Feedback Round	Review feedback, adjust if needed
24.06.2026	Final Presentation	Live demo + report submission via Moodle

5.5 Weekly Status Checklist

- ☐ **Week 1 (13–19.05):** WP0 setup complete
- ☐ **Week 2 (20–26.05):** WP1 data integration done, Proposal prepared
- ☐ **Week 3 (27.05–02.06):** WP2 data cleaning complete, WP3 started
- ☐ **Week 4 (03–09.06):** WP3 features complete, WP4 preprocessing ready
- ☐ **Week 5 (10–16.06):** WP5 all 5 models trained and validated
- ☐ **Week 6 (17–23.06):** WP6 best model selected, WP7 Shiny App deployed
- ☐ **Week 7 (24.06):** Final presentation & report submitted

6 Variable selection and Data Cleaning

After all the data has been mapped put in a database we construct the final selection of variables used for modelling and clean this data.

```
library(DBI)
library(RPostgres)
```

Warning: package 'RPostgres' was built under R version 4.5.3

```
library(dotenv)
```

Warning: package 'dotenv' was built under R version 4.5.2

```
library(here)
```

here() starts at D:/Documents/00-Personal/01-FHTW/SS2026/MAD-BB-2-SS2026-SOE-R/99-Project-SolutionEngineering/SOE-R-House-Price-Prediction

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union


```
library(tidyr)
library(forcats)
library(stringr)

load_dot_env(here::here(".env"))
```

```
con <- dbConnect(
  RPostgres::Postgres(),
  host      = Sys.getenv("DB_HOST"),
  port      = as.integer(Sys.getenv("DB_PORT")),
  dbname    = Sys.getenv("DB_NAME"),
  user      = Sys.getenv("DB_USER"),
  password  = Sys.getenv("DB_PASSWORD")
)
dbIsValid(con)
```

```
[1] TRUE
```

6.1 Subset the Data

After mapping the data out and looking at the variable description, we limit the potentially used features. These features should be ones, that a homeowner likely knows about their house.

Restrict the features to: - land square feet - waterfront type - view quality - electricity availability - water availability - street type - coordinates (longitude, latitude)

```
df_1 <- dbGetQuery(con, '
  SELECT parcel_no, appraisal_account_type, buildings, land_net_squarefeet, waterfront_type,
  utility_electric, utility_water, street_type, latitude, longitude
  FROM "APPRAISAL_ACCOUNT"
  WHERE appraisal_account_type = \'Residential\'
')
```

We get property specifics: - condition - number of stories - number of bedrooms - number of bathrooms - units - year built - year remodeled

```
df_2 <- dbGetQuery(con, "
  SELECT i.parcel_no, i.building_id, i.property_type, i.condition,
         b.stories, b.bedrooms, b.bathrooms, b.units, b.year_built, b.year_remodeled
  FROM \"IMPROVEMENT\" i
  LEFT JOIN \"IMPROVEMENT_BUILT_AS\" b
    ON i.parcel_no = b.parcel_no
    AND i.building_id = b.building_id
  WHERE i.property_type = 'Residential'
")
```

We get the current property value.

```
df_3 <- dbGetQuery(con, '
  SELECT parcel_no, total_market_value_current_year
  FROM "TAX_ACCOUNT"
')
```

we join the data together.

```
df_all <- df_1 %>%
  left_join(df_2, by = "parcel_no") %>%
  left_join(df_3, by = "parcel_no")
```

6.1.1 Cleaning

We obtain the dataset with selected features. Now we go ahead to prepare the data for initial exploratory data analysis.

```
df_all <- df_all %>%
  select( -property_type)
```

We first select lands with buildings on it, as we are interested in residential home valuations.

```
df_all <- df_all %>%
  filter(buildings != 0)
```

Then we look at the missing values and treat them one-by-one.

```
colSums(is.na(df_all))
```

parcel_no	appraisal_account_type
0	0
buildings	land_net_squarefeet
0	3
waterfront_type	view_quality
256540	252509
utility_electric	utility_water
0	0
street_type	latitude
0	5
longitude	building_id
5	29816
condition	stories
29816	29846
bedrooms	bathrooms
29846	29846
units	year_built
29887	29846
year_remodeled	total_market_value_current_year
42869	0

We remove observations that have no bedroom or bathroom recorded as these would be one of the features easiest for the user to input and we want to use them. If they are missing it likely hints at a data recording issue or a building type that is not a classical residential home (which is out of scope for this project).

```
df_all <- df_all %>%
  filter(!is.na(bathrooms))
```

We also remove the 5 observations missing coordinates or square footage.

```
df_all <- df_all %>%
  filter(!is.na(latitude) & !is.na(longitude) & !is.na(land_net_squarefeet))
```

Waterfront type records River, Lake, Stream/Creek or Salt. We fill in missing values to represent no waterfront, as such category does not exist and missing specification likely represents that the property has no direct water connection.

```
df_all <- df_all %>%
  mutate(waterfront_type = replace_na(waterfront_type, "None"))
unique(df_all$waterfront_type)
```

```
[1] "WF Salt"          "None"             "WF Lake"          "WF Stream/Creek"
[5] "WF River"
```

View quality currently has 9 categories that is slightly excessive, and more confusing for the potential user and increases model complexity.

```
unique(df_all$view_quality)
```

```
[1] NA                "View Lim"         "View Lim -"       "View Lim +"
[5] "View Avg +"      "View Avg"         "View Good"        "View Good +"
[9] "View V-Good"     "View V-Good +"
```

We collapse the categories into “limited”, “average”, “good”, “very good”. We replace NAs with uncategorized.

```
df_all <- df_all %>%
  mutate(
    view_quality = replace_na(view_quality, "uncategorized"),

    view_quality = fct_collapse(view_quality,
      "limited"    = c("View Lim", "View Lim -", "View Lim +"),
      "average"   = c("View Avg", "View Avg +"),
      "good"      = c("View Good", "View Good +"),
      "very good" = c("View V-Good", "View V-Good +")
    )
  )
unique(df_all$view_quality)
```

```
[1] uncategorized limited      average      good          very good
Levels: uncategorized average good limited very good
```

We order the view types.

```
df_all$view_quality <- factor(df_all$view_quality,
                             levels = c("uncategorized", "limited", "average", "good", "very good"),
                             ordered = TRUE)
```

For the year remodeled, we will proceed as no remodeling happened, if the year is not recorded.

```
df_all <- df_all %>%
  mutate(year_remodeled = replace_na(year_remodeled, 0))
```

Lastly, in the units feature we convert the few missing values to 1 unit after looking into the data manually, and also convert 0 units to 1 as the empirical meaning of having no separate units and the whole unit being one (for example family home) is the same.

```
df_all <- df_all %>%
  mutate(units = if_else(is.na(units) | units == 0, 1, units))
```

We have now the complete dataset with the preselected features.

```
colSums(is.na(df_all))
```

parcel_no	appraisal_account_type
0	0
buildings	land_net_squarefeet
0	0
waterfront_type	view_quality
0	0
utility_electric	utility_water
0	0
street_type	latitude
0	0
longitude	building_id
0	0
condition	stories
0	0
bedrooms	bathrooms
0	0
units	year_built
0	0
year_remodeled	total_market_value_current_year
0	0

A few post checks we should still apply.

```
unique(df_all$utility_electric)
```

```
[1] "POWER INSTALLED"      "POWER AVAILABLE"      "POWER NO - COMMENT"
```

```
df_all <- df_all %>%  
  mutate(utility_electric = str_replace(utility_electric, "POWER NO - COMMENT", "POWER NO"))  
unique(df_all$utility_electric)
```

```
[1] "POWER INSTALLED" "POWER AVAILABLE" "POWER NO"
```

```
unique(df_all$utility_water)
```

```
[1] "WATER INSTALLED" "WATER NO"          "WATER AVAILABLE"
```

```
unique(df_all$street_type)
```

```
[1] "STREET UNPAVED" "PAVED"             "STREET NO ROAD"
```

Check coordinate range.

```
df_all %>%  
  summarise(  
    min_lat = min(latitude, na.rm = TRUE),  
    max_lat = max(latitude, na.rm = TRUE),  
    min_lon = min(longitude, na.rm = TRUE),  
    max_lon = max(longitude, na.rm = TRUE)  
  )
```

```
      min_lat max_lat min_lon max_lon  
1 46.73973 47.40335 -122.8406 -121.4681
```

```
unique(df_all$condition)
```

```
[1] "Average"      "Extra Poor"      "Uninhabitable" "Poor"  
[5] "Fair"         "Very Poor"
```

```
df_all$condition <- factor(df_all$condition,
                           levels = c("Uninhabitable",
                                       "Extra Poor",
                                       "Very Poor",
                                       "Poor",
                                       "Fair",
                                       "Average"),
                           ordered = TRUE)
```

```
df_all %>%
  select(stories, bedrooms, bathrooms) %>%
  slice_max(order_by = stories, n = 5)
```

	stories	bedrooms	bathrooms
1	15	3	2.00
2	10	3	2.00
3	8	0	0.50
4	4	3	3.75
5	4	3	4.00

The top 3 observation are very unlikely with this combination of bedrooms and bathrooms, so we remove them.

```
to_remove <- df_all %>%
  slice_max(order_by = stories, n = 3, with_ties = FALSE)

df_all <- df_all %>%
  anti_join(to_remove)
```

```
Joining with `by` = join_by(parcel_no, appraisal_account_type, buildings,
land_net_squarefeet, waterfront_type, view_quality, utility_electric,
utility_water, street_type, latitude, longitude, building_id, condition,
stories, bedrooms, bathrooms, units, year_built, year_remodeled,
total_market_value_current_year)`
```

Looking at the data again manually, and in particular the combination of bathrooms and bedrooms with teh units we also detect some anomalies that we remove.

```
df_all <- df_all %>%
  filter(bathrooms <= 17 & bedrooms <= 21)
```

The year remodeled being 0 means no remodeling has been recorded on the house.

```
df_all %>%
  summarise(
    min_built      = min(year_built, na.rm = TRUE),
    max_built      = max(year_built, na.rm = TRUE),
    min_remodeled  = min(year_remodeled, na.rm = TRUE),
    max_remodeled  = max(year_remodeled, na.rm = TRUE)
  )
```

```
   min_built max_built min_remodeled max_remodeled
1    1861    2026         0         2026
```

We additionally check if the year remodeled is not 0 it is always more recent than the year built.

```
impossible_dates <- df_all %>%
  filter(year_remodeled > 0 & year_remodeled < year_built)

nrow(impossible_dates)
```

```
[1] 0
```

Check the valuation range.

```
df_all %>%
  summarise(
    min_value      = min(total_market_value_current_year, na.rm = TRUE),
    max_value      = max(total_market_value_current_year, na.rm = TRUE)
  )
```

```
   min_value max_value
1     8400 10627600
```



```
#save(df_all, file = here::here("data", "df_all.RData"))
```

References